An Input-Regime Audit of Conflict Detection for Retrieval-Augmented Generation

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ABSTRACT

Vector-database-backed retrieval-augmented generation (RAG) pipelines increasingly add a *conflict-detection* stage that flags contradictions among retrieved passages before generation. Two recent directions dominate practice. The first is a pairwise NLI cross-encoder over retrieved passages, the standard baseline detector across conflict-aware RAG systems and the stack deployed by ConflictRAG [7]. The second is a *query-conditioned* formulation that lets the NLI label depend on the question being answered, recently benchmarked by Canby et al. [1]. Practitioners must choose between them with little empirical guidance. We provide that guidance. We audit three detectors: the classic DeBERTa-MNLI cross-encoder, an LLM-as-judge baseline, and the same LLM judge question-conditioned. We evaluate them on curated probes (n=63 with bootstrap CIs), a 10-model LLM sweep, a prompt-robustness ablation, a long-passage realism step, an answer-bearing scoping study, and external validation on 100 human-labelled FEVER pairs. Five findings guide deployment. (1) LLM judges retain perfect precision on labelled FEVER pairs and flag far fewer top-k retrieved non-gold passages than the cross-encoder (0.09 vs 0.43 flag rate). (2) A same-name "sense-trap" false positive systematically fools both the cross-encoder and pairwise LLM judging. Only question-conditioning closes the gap (precision 0.68 $\rightarrow$ 1.00 on curated probes, replicated for 8 of 10 LLM families). (3) The cross-encoder vs LLM trade is *input-distribution dependent*. The cross-encoder is competitive on its native short claim/evidence distribution (FEVER F1 0.85) but unusable on long multi-claim passages (F1 0.18). (4) The small-model regression under the question-conditioning prompt is a capability limit, not a format limit. We verify this by inspecting raw outputs. (5) Pairwise conflict detection is ill-posed on multi-hop evidence fragments and must be scoped to answer-bearing units. We synthesise the findings into a five-row input-regime ladder. The ladder recommends a detector for each retrieval context a vector-DB practitioner is likely to encounter. Code and result artefacts are released.

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The source code, data, and/or other artifacts have been made available at https://github.com/sarkar-dipankar/ebrag-vecdb-2026-paper.

1 INTRODUCTION

A vector-database-backed RAG pipeline returns a top-k set of passages. A growing class of *conflict-aware* systems then runs a *conflict-detection* stage on those passages before generation [3, 5, 8–11]. The detector's signal gates downstream choices: multi-agent debate over conflicting sources [9], source-reliability consolidation [8], structured support/refute output [6], or calibrated abstention via standard confidence calibration [4]. The detector is the load-bearing component upstream of all of these: a poor detector silently caps every method built on top of it.

Two recent directions dominate practice. A pairwise NLI cross-encoder applied to retrieved-passage pairs is the standard baseline detector across conflict-aware pipelines [3, 7, 9]. As a recent conflict-aware RAG system, ConflictRAG [7] reports 88.7% F1 and downstream correctness gains of 5.3 to 6.1% over conflict-aware baselines. Concurrent work by Canby et al. [1] formalises and benchmarks *query-conditioned NLI*: the entailment label depends on a premise, a hypothesis, *and* a query that specifies the aspect under comparison. A practitioner picking a detector for a vector-DB pipeline needs to know three things: which approach, on what kind of input, and with what model.

We provide that empirical guidance. We audit a representative pairwise NLI cross-encoder (DeBERTa-base-MNLI), an LLM-as-judge baseline in the style of Gokul et al. [3], and the same LLM judge question-conditioned in the style of Canby et al. [1]. The setting is the conflict-aware RAG pipeline of ConflictRAG [7] and adjacent systems. We evaluate the three detectors across a curated probe set, a 10-model LLM sweep, a small-model prompt-robustness ablation, a long-passage realism study, an answer-bearing scoping experiment, and external validation on 100 human-labelled FEVER claim/evidence pairs. We report bootstrap 95% confidence intervals throughout, so the audit's orderings are statistically interpretable.

Where this sits in the vector-DB pipeline. The detector we audit is a single stage in a longer vector-DB-backed RAG pipeline:

top-k retrieval $\rightarrow$ passage units $\rightarrow$ pair construction $\rightarrow$ detector $\rightarrow$ generation policy

The vector-DB practitioner controls five design choices that change what the detector sees: (i) chunk size and granularity (sentence, paragraph, multi-hop); (ii) top-k and how many pairs are checked; (iii) whether pairs are formed sentence-level, paragraph-level, or only on *answer-bearing* units; (iv) whether conflict detection runs before or after a question-conditioned extraction step; and (v) detector cost and latency vs precision. Our audit holds the surrounding

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pipeline fixed and varies the detector. The findings below land on choices (i), (iii), (iv), and (v).

Contributions.

(1) **A mechanistic failure mode of pairwise judging: the sense trap.** Same-name-different-referent passage pairs ("Paris is the capital of France" vs. "Paris is a town in Texas") systematically fool both the pairwise NLI cross-encoder baseline used across conflict-aware RAG [3, 7, 9] and a pairwise LLM judge. We isolate the mechanism: pairwise judging lacks the question that says *which* referent is relevant.
(2) **An empirical audit of the query-conditioning fix.** Conditioning the LLM judge on the question is the operational form of the query-conditioned NLI proposed by Canby et al. [1]. It closes the sense-trap gap on the curated set (precision 0.68 → 1.00) and replicates across 8 of 10 open-weight LLM families (§6) with non-overlapping bootstrap CIs.
(3) **Capability, not format.** The small-Gemma regression under the question-conditioning prompt is a reasoning limit, not a JSON-format artefact. Raw model output inspection shows it (§7).
(4) **Scoping: pairwise conflict is ill-posed on multi-hop evidence fragments.** An evidence paragraph about one entity ("Scott Derrickson is American") does not contradict a relationship verdict ("Derrickson and Wood are of different nationalities"). The judge's `neutral` is correct. The audit must be scoped to answer-bearing pairs. This is a methodological clarification useful to any downstream vector-DB pipeline that retrieves multi-hop evidence.
(5) **The detector trade is input-regime dependent.** The cross-encoder is competitive on its native short claim/evidence distribution (FEVER F1 0.85) but unusable on long multi-claim passages (F1 0.18). LLM judges are robust across lengths but more conservative on recall. We chart the five-row regime ladder (Table 8) as a deployment guide.

Relation to prior work. Canby et al. [1] formalise query-conditioned NLI as a benchmark task; we instead audit it as a *deployable conflict-detection signal* for RAG. We characterise where its fix dominates the pairwise alternative and identify the model-capability regime in which it backfires. Wang et al. [7] build a two-stage conflict-aware RAG pipeline (embedding-based MLP classifier with selective LLM refinement, then source-credibility weighting) and report end-task gains; we audit the underlying pairwise NLI detector signal in isolation, upstream of any source-credibility stage. Dhole et al. [2] and Wang et al. [9] target the generator's handling of conflicting evidence; we work on the detector that supplies their input signal.

2 RELATED WORK

Conflict detection on top of retrieval. ConflictRAG [7] is the most direct system motivation for our audit. It introduces a two-stage conflict-detection module (an embedding-based MLP classifier with selective LLM refinement) followed by source-credibility weighting, reporting 88.7% conflict-detection F1 and 5.3 to 6.1% downstream correctness gains over conflict-aware baselines. Our audit complements that system result by isolating the pairwise NLI signal, the standard cross-encoder baseline used across this literature [3, 9], and characterising where it succeeds and where it fails. Gokul et al. [3] evaluate LLMs as context-validators in RAG pipelines, a direct counterpart to our `llm_bare` detector. Zeng et al. [11] construct a fact-checking dataset for RAG robustness against misleading retrieval, complementing the FEVER claim/evidence audit we run in §10.

Query-conditioned NLI. Canby et al. [1] introduce and benchmark query-conditioned NLI. The entailment/contradiction label is conditioned on a query alongside the premise and hypothesis. Their contribution is the formalisation and a benchmark. Ours is an *empirical deployment audit* of the same idea for RAG. We ask how the question-conditioned variant behaves vs. pairwise NLI across model families (§6), input lengths (§8, §9), and real human-labelled retrieved evidence (§10). We also identify the small-model capability regime in which the fix regresses (§7).

Generator-side conflict-aware RAG. A line of work targets how the *generator* handles disagreement: multi-agent grounded debate (Madam-Rag + RAMDocs) [9]; source-reliability consolidation [8]; structured support/refute output [6]; falsification for sycophancy [5]; the prior-vs-context tug-of-war [10]; stance-diversified retrieved evidence for argumentation [2]. All of these consume a detector signal and reason over conflicting inputs; none audit the detector itself. We operate strictly upstream, measuring where pairwise detection (cross-encoder NLI and LLM-as-judge alike) fails and where question-conditioning closes the gap, so the downstream choices these systems make rest on a characterised signal.

NLI as a building block. Cross-encoder NLI predates the RAG conflict-detection literature. We use DeBERTa-base-MNLI as the canonical pairwise NLI representative because variants of this model family are the standard baseline detector across conflict-aware RAG [3, 7, 9]. Temperature scaling and calibration of NLI heads [4] are orthogonal to the audit. We report contradiction-class probabilities directly.

3 DETECTORS

We compare three detectors. All are evaluated as a binary contradiction classifier.

cross_encoder_nli. DeBERTa-base-MNLI, the "classic" pairwise detector used by many conflict-aware RAG pipelines. Input: a single text-pair string. Pairwise by design, with no slot for the question.

llm_bare. An LLM judge prompted to label the pair entailment / contradiction / neutral. Pairwise like the cross-encoder, but with arbitrary instruction-following.

llm_answer_relevant. The same LLM judge prompted to decide whether passages A and B give *conflicting answers to a specific question.* The question enters the prompt; A and B are otherwise the same texts as in the bare variant. This is the operational form of the query-conditioned NLI of Canby et al. [1], used here as a deployable conflict-detection signal.

Detector	Precision (95% CI)	F1	Sense-trap acc (95% CI)
cross-enc NLI	0.68 [0.52,0.83]	0.81	0.67 [0.38,0.92]
LLM bare	0.90 [0.77,1.00]	0.95	0.75 [0.46,1.00]
LLM answer-rel	1.00 [1.00,1.00]	1.00	1.00 [1.00,1.00]

Table 1: Detector ladder on the curated probe set ($n = 63$). The cross-encoder is structurally unable to be question-conditioned; only the answer-relevant LLM judge closes the sense-trap gap.

4 PROBES AND DATA

Curated probe set. 63 probes spanning 9 categories: same-name sense traps, explicit conflicts, numeric/temporal/implicit conflicts, paraphrase agreement, unit equivalence, negation agreement, topic shift. Gold-correctness invariants are unit-tested (every conflict category is gold=conflict; every no-conflict category is gold=no-conflict; template-generated numeric/temporal/unit pairs differ in only the controlled value).

Long answer-bearing pairs. For 14 single-hop factoid questions with two competing answers, we LLM-write 3 to 4 sentence passages that each assert a specific answer. This gives conflict, agreement, and topic-shift pairs over long real-sounding text.

FEVER (real, human-labelled). 100 VERIFIABLE pairs from `copenlu/fever_gold_evidence`: 50 REFUTES (conflict) and 50 SUPPORTS (no-conflict). Premise is the gold evidence sentence and hypothesis is the claim.

5 THE SENSE-TRAP AND ANSWER-RELEVANCE

On the curated probes, a same-name-different-referent pair ("Paris, France" / "Paris, Texas" with question *What is the capital of France?*) is consistently flagged as a contradiction by the cross-encoder and frequently by the bare LLM judge. Both decide pairwise without the question and default to the assumption that the two passages are about the same "Paris". Conditioning the judge on the question ("Do A and B give conflicting answers to *this* question?") removes the failure without recall loss on the genuine conflicts: precision rises from 0.68 to 1.00 on the curated probes (Table 1), and the cross-encoder's upper CI (0.83) sits below answer-relevance's point estimate (1.00).

6 CROSS-MODEL SCALING

We run the bare vs. answer-relevant comparison across the 10-model `DEFAULT_SWEEP` (six families, 4B to 671B parameters) with bootstrap precision CIs (Table 2). Across families, the four most capable models (gpt-oss:20b, gpt-oss:120b, qwen3-next:80b, glm-4.6) reach perfect precision under the answer-relevance prompt with bare CIs that sit clearly below. 8 of 10 models improve. The two small Gemma models regress with non-overlapping CIs: gemma3:4b collapses to P 0.51 $[0.37, 0.65]$ (FP 1→25). The mean lift is small (+0.011) because the two regressions drag it down; the headline is the family-spanning capability dependence, not the mean.

The ladder against a fixed cross-encoder. The cross-encoder NLI is model-fixed (DeBERTa-MNLI), so its precision on the curated set is a constant reference. The per-LLM-model picture against it is in Table 3. **Bare LLM judging beats the cross-encoder for every one of the 10 LLM models** on this probe set. Answer-relevance beats it for 8 of 10, and the full ladder (answer-rel ≥ bare ≥ cross-encoder) holds for the same 8 models that improved under answer-relevance in Table 2. The two small Gemma regressors are the *only* models on which an LLM judge would lose to the DeBERTa-MNLI cross-encoder under the answer-relevant prompt: the binding constraint is the capability limit (§7), not the framing.

7 CAPABILITY, NOT FORMAT

The small-Gemma regression could be a JSON-format failure or a reasoning limit. We disentangle (Table 4) by judging under three answer-relevance prompts: the production JSON prompt, a one-word `plain` variant, and a `fewshot` variant with two worked examples. We test all three on gemma3:4b, gemma3:12b, and the capable gpt-oss:20b control. We then *inspect the raw outputs* on the sense-trap false positives. gemma3:4b emits clean, parseable JSON that often verbalises the right distinction. On Mercury planet vs. element, its rationale reads "Passage A answers the question, while Passage B discusses a chemical element". The model still labels the pair `contradiction`. This is a decision/reasoning failure, not a format artefact. Switching to one-word output only shifts the threshold (FP 25 → 11), and the model still fails on Paris/Cambridge sense traps. Few-shot demonstration partially rescues gemma3:12b (sense-trap acc 0.50 → 0.67) but not gemma3:4b. gpt-oss:20b is robust across all variants and goes to perfect with few-shot.

8 REALISM: SCOPING TO ANSWER-BEARING PAIRS

A first realism run with HotpotQA-distractor produced an honest negative. Pairwise conflict recall stayed low (≤ 0.45) even after we fixed the refutation generator (boolean-flip opposite-answer refutations). On this regime ($n = 59$ pairs from the corrected refutation generator), the cross-encoder hits F1 0.18, LLM-bare 0.62, and the question-conditioned judge 0.26. Manual inspection clarified the cause. A HotpotQA "supporting" paragraph is an *evidence fragment about one entity* ("Scott Derrickson is an American director"). An opposite-answer refutation is a *multi-hop relationship verdict* that often *agrees on the shared fact* ("born in Denver, Colorado" is still American) and differs only on the combined conclusion. S and R therefore *do not contradict pairwise*. The judge's `neutral` is correct, and the gold is ill-posed. Pairwise conflict detection is only well-defined when *both* passages assert an answer to the same question. We accordingly scope the experiments to answer-bearing units.

9 LONG ANSWER-BEARING PAIRS

Restricting to 14 single-hop factoid questions, we LLM-write a 3 to 4 sentence passage asserting each candidate answer and build conflict / agreement / topic-shift pairs over the resulting long text (Table 5). Conflict recall recovers across detectors (cross-encoder 0.05 → 0.50 versus multi-hop evidence; LLM-bare 0.45 → 0.86; answer-relevant 0.15 → 0.79). When the gold is well-posed and both sides are answer-bearing, detectors detect. On long text, the LLM judges dominate F1 (0.89 vs 0.67). The cross-encoder regains perfect precision but length caps its recall at 0.50. Answer-relevance

Model	Family	Size (B)	bare P (95% CI)	answer-rel P (95% CI)	Lift	FP b⟶ar
`gpt-oss:20b`	gpt-oss	20	0.84 [0.70,0.96]	1.00 [1.00,1.00]	+0.16	5⟶0
`gpt-oss:120b`	gpt-oss	120	0.90 [0.77,1.00]	1.00 [1.00,1.00]	+0.10	3⟶0
`gemma3:4b`	gemma	4	0.96 [0.88,1.00]	0.51 [0.37,0.65]	-0.45	1⟶25
`gemma3:12b`	gemma	12	0.76 [0.62,0.90]	0.63 [0.47,0.78]	-0.13	8⟶15
`gemma3:27b`	gemma	27	0.83 [0.69,0.96]	0.87 [0.73,0.97]	+0.03	5⟶4
`qwen3-next:80b`	qwen	80	0.93 [0.81,1.00]	1.00 [1.00,1.00]	+0.07	2⟶0
`deepseek-v3.1:671b`	deepseek	671	0.84 [0.70,0.95]	0.87 [0.73,0.97]	+0.03	5⟶4
`glm-4.6`	glm	–	0.84 [0.70,0.96]	1.00 [1.00,1.00]	+0.16	5⟶0
`ministral-3:8b`	mistral	8	0.70 [0.56,0.84]	0.74 [0.59,0.88]	+0.04	11⟶9
`nemotron-3-nano:30b`	nemotron	30	0.87 [0.73,0.97]	0.96 [0.88,1.00]	+0.10	4⟶1

Table 2: Answer-relevance precision lift across the 10-model sweep (8/10 improve; mean lift +0.011). The four most capable models reach perfect precision; the two small Gemma models regress (non-overlapping CIs). $n = 63$ **probes.**

LLM model	cross-enc P	LLM-bare P	LLM-ar P	ladder
`gpt-oss:20b`	0.68	0.84	1.00	✓
`gpt-oss:120b`	0.68	0.90	1.00	✓
`gemma3:4b`	0.68	0.96	0.51	×
`gemma3:12b`	0.68	0.76	0.63	×
`gemma3:27b`	0.68	0.83	0.87	✓
`qwen3-next:80b`	0.68	0.93	1.00	✓
`deepseek-v3.1:671b`	0.68	0.84	0.87	✓
`glm-4.6`	0.68	0.84	1.00	✓
`ministral-3:8b`	0.68	0.70	0.74	✓
`nemotron-3-nano:30b`	0.68	0.87	0.96	✓

Table 3: Per-model ladder against the fixed cross-encoder reference (DeBERTa-MNLI, $P = 0.68$ **on** $n = 63$**). Bare LLM judging beats the cross-encoder for 10/10 models; answer-relevance beats it for 8/10; the full ladder (answer-rel ≥ bare ≥ cross-enc) holds for 8/10. The two small Gemma regressors drop *below* the cross-encoder under answer-relevance, making them the only models where a cross-encoder would beat the LLM judge.**

Model	Variant	Precision	F1	Sense-trap acc	FP
`gemma3:4b`	json	0.51	0.68	0.00	25
`gemma3:4b`	plain	0.67	0.75	0.58	11
`gemma3:4b`	fewshot	0.62	0.76	0.42	16
`gemma3:12b`	json	0.63	0.78	0.50	15
`gemma3:12b`	plain	0.63	0.78	0.33	15
`gemma3:12b`	fewshot	0.76	0.87	0.67	8
`gpt-oss:20b`	json	0.96	0.98	0.92	1
`gpt-oss:20b`	plain	0.93	0.96	0.83	2
`gpt-oss:20b`	fewshot	1.00	1.00	1.00	0

Table 4: Prompt-robustness ablation ($n = 63$**). The small-Gemma regression is a *capability* limit, not a format artefact: gemma3:4b emits clean parseable JSON that verbalises the right distinction and still mislabels (manual inspection); switching to a single-word format only shifts its threshold (FP** $25 \rightarrow 11$**); few-shot demonstration partially rescues gemma3:12b but not gemma3:4b.**

has perfect precision at a small recall cost. Its big sense-trap precision win is muted here because these topic-shift distractors are different-*subject*, not same-name.

10 EXTERNAL VALIDATION ON REAL FEVER

We sanity-check the previous result on real human-labelled data. `copenlu/fever_gold_evidence` bundles claim + label (SUPPORTS / REFUTES) + gold evidence sentences. We evaluate the three detectors on 100 VERIFIABLE pairs balanced 50/50 (Table 6). The LLM judges keep **perfect precision** (1.00 [1.00, 1.00]) on real data, so the precision finding survives external validation. The cross-encoder, on its native training distribution (short claim/evidence NLI), is *competitive on F1* (0.85 vs 0.84/0.81) via higher recall (0.80 vs 0.72/0.68). Answer-relevance neither helps nor hurts here (no sense-trap failures to fix on FEVER pairs), consistent with the answer-bearing result.

Detector	Precision (95% CI)	Recall (95% CI)	F1
cross_encoder_nli	1.00 [1.00,1.00]	0.50 [0.25,0.78]	0.67
llm_bare	0.92 [0.75,1.00]	0.86 [0.65,1.00]	0.89
llm_answer_relevant	1.00 [1.00,1.00]	0.79 [0.55,1.00]	0.88

Table 5: Long answer-bearing passage pairs ($n = 42$: 14 conflict / 14 agreement / 14 topic-shift, all 3–4 sentence LLM-written passages asserting a specific answer). LLM judges dominate F1; the cross-encoder regains perfect precision but length caps recall.

Detector	Precision (95% CI)	Recall (95% CI)	F1
cross_encoder_nli	0.91 [0.82,0.98]	0.80 [0.68,0.91]	0.85
llm_bare	1.00 [1.00,1.00]	0.72 [0.59,0.84]	0.84
llm_answer_relevant	1.00 [1.00,1.00]	0.68 [0.53,0.81]	0.81

Table 6: External validation on real human-labelled FEVER pairs ($n = 100$; `copenlu/fever_gold_evidence`). On short claim–evidence NLI, the cross-encoder's native distribution, it is competitive; LLM judges retain perfect precision (no false positives on real data) but are more conservative on recall.

11 VECTOR-RETRIEVAL REALISM

The experiments above evaluate the three detectors on curated or pre-paired data. This section closes the deployment loop. We embed all 100 FEVER gold evidence sentences with `all-MiniLM-L6-v2` and use exact cosine retrieval as a deterministic small-corpus proxy for the vector-index stage, sidestepping ANN variance in this audit. We retrieve top-$k = 5$ passages for each of the 100 claims. Each retrieved (claim, passage) pair is then scored by all three detectors. This is the pipeline a vector-DB practitioner actually runs: dense top-k followed by pairwise conflict-checking on the returned passages.

Retrieval works as expected. Gold evidence is in the top-5 for 0.87 of claims (recall@5) and at rank 1 for 0.76 (recall@1). The detector behaviour on the *retrieved* pairs is the interesting part (Table 7). The cross-encoder flags conflict on 0.43 [0.39, 0.48] of retrieved pairs. The two LLM judges flag 0.09 and 0.08. Most retrieved passages for a given claim are topically related but not contradicting (the same Wikipedia article cluster), so a high flag rate here is a potential over-flagging surface in deployment, not a true detection win. On the subset where the retrieved passage is the FEVER gold evidence (where the SUPPORTS/REFUTES label gives a per-pair ground truth), all three detectors are within a few points of each other on precision-on-gold (0.85, 0.93, 0.89), so the cross-encoder's extra flags are coming from the non-gold retrieved pairs, not from a better gold signal. Latency separates the detectors by roughly 30× (0.16 s for the cross-encoder, 4 to 5 s for the LLM judges). The cost ladder and the over-flagging ladder both push the same way on long real retrieved pairs as they did on the synthetic answer-bearing pairs (§9) and the FEVER claim/evidence pairs (§10).

Detector	Flag rate (95% CI)	Prec@gold	Latency (s)
cross-enc NLI	0.43 [0.39,0.48]	0.85	0.16
LLM bare	0.09 [0.06,0.11]	0.93	4.05
LLM answer-rel	0.08 [0.06,0.11]	0.89	5.21

Table 7: Vector-retrieval realism. Top-k dense retrieval ($k = 5$, exact cosine over an embedded corpus of $n = 100$ FEVER evidence sentences, all-MiniLM-L6-v2 embeddings) for $n = 100$ FEVER claims; gold-evidence recall@k is 0.87 (recall@1 0.76). Each retrieved (claim, passage) pair is scored by the three detectors. The cross-encoder's higher flag rate at comparable precision-on-gold reflects the same precision/recall trade seen on synthetic and FEVER pairs (§9, §10). Per-detector mean latency shows the cost ladder a vector-DB practitioner faces in deployment.

Input regime	Best detector	Why
Short curated, sense traps present (§5)	LLM_ANSWER_RELEVANT (P 1.00)	only it is question-conditioned
Short real claim–evidence (FEVER, §10)	cross-enc ≈ LLM_BARE on F1	cross-encoder's home turf
Real top-k retrieved pairs (§11)	LLM judges (flag 0.09 vs 0.43)	cross-enc over-fires on non-gold
Long answer-bearing passages (§9)	LLM judges (F1 0.89)	cross-encoder length-capped
Long multi-hop fragments (§8)	ill-posed; scope to answer-bearing	pairwise conflict undefined

Table 8: Detector choice depends on input regime. The LLM advantage is conditional on sense-trap presence and passage length; the cross-encoder remains useful for short native-distribution NLI.

12 SYNTHESIS: AN INPUT-REGIME LADDER

The detector choice is not a single ladder but an *input-regime* choice (Table 8). The LLM-judge advantage is conditional on (a) the presence of same-name / sense-trap failures (where answer-relevance dominates) and (b) passage length (where the cross-encoder breaks down). On the cross-encoder's short native NLI distribution, it is competitive. The practical recommendation: use a question-conditioned LLM judge on *answer-bearing* units (single-hop / claim-level). The cross-encoder is appropriate only for short native-distribution pairs where high recall matters more than the sense-trap precision risk.

Implications for vector-DB practitioners. A practitioner building a conflict-aware RAG pipeline on top of a vector store typically faces three concrete choices: which detector to deploy, on what unit of retrieved content, and how to interpret the resulting signal under model heterogeneity. The input-regime ladder (Table 8) gives a direct recommendation for each. For pipelines that retrieve and conflict-check *short claim/evidence pairs* (e.g., fact-checking RAG and similar pairwise filters), the cross-encoder is the right default. It matches the LLM judge on F1 with much lower compute, and the sense-trap precision risk is small at sentence scale. For pipelines that retrieve *paragraph-length* passages directly into the conflict-detection stage, the cross-encoder fails (F1 0.18 at our long-passage scale) and an LLM judge is required. Question-conditioning the LLM judge is then the precision-maximising default, unless the deployed LLM is small enough to fall into the capability regime (Table 4). For pipelines that retrieve *multi-hop evidence fragments* (HotpotQA-style or graph-RAG-style sub-evidence), pairwise conflict detection is ill-posed on the retrieved units themselves. Conflict-aware logic should instead be deferred to a downstream synthesis step that operates on answer-bearing representations (§8).

13 LIMITATIONS

We group the audit's limitations by the dimension they affect.

Probe set scale and validation. The curated probe set is n=63 and was built by a single author with no independent inter-annotator agreement. Bootstrap CIs make the orderings statistically clean, but the absolute gaps need a larger, held-out-author probe set to tighten. FEVER (n=100) provides the real-corpus external check.

Coverage of input regimes. The long answer-bearing passages are LLM-written. FEVER covers short claim/evidence pairs on real text but not long answer-bearing text on real corpora; closing that specific gap would need a hand-labelled long-passage benchmark. The retrieval-realism check (§11) uses a 100-passage embedded corpus, sufficient for a deployment sanity check on flag rate and latency but not for realistic vector-DB scale.

Scope of what the audit varies. The cross-encoder is fixed at DeBERTa-base-MNLI as the canonical pairwise NLI representative used across conflict-aware RAG [3, 7, 9]; larger NLI models may shift the input-regime ladder. The 10-model LLM sweep uses one prompt per condition, and the prompt-robustness study (§7) covers only three variants on three models, so prompt-engineering rescue of small models is plausibly under-explored.

Downstream effect on RAG generation. We measure detector behaviour, not the downstream effect of detector choice on RAG generation quality. A natural follow-up is to propagate the sense-trap precision win into end-task faithfulness and abstention numbers.

Concurrent work. Query-conditioned NLI was concurrently formalised as a benchmark by Canby et al. [1]. Our contribution is the deployment audit, not the original technique; the audit is complementary to and extends their benchmark.

14 CONCLUSION

We audit conflict detection on top of vector retrieval for RAG. Pairwise NLI is a brittle signal: a same-name sense trap systematically false-positives, and the failure is mechanistic (a missing question slot), not a model-capacity issue. The question-conditioned formulation of Canby et al. [1] closes the gap when deployed as an LLM-judge detector, and the closure replicates across 8 of 10 open-weight families with non-overlapping CIs.

The choice between the pairwise NLI cross-encoder commonly used in conflict-aware RAG [3, 7, 9] and an LLM judge is *not a global recommendation but an input-regime one.* The cross-encoder wins on short claim/evidence pairs and is unusable on long passages. LLM judges are robust across lengths but conservative on recall. Small models cannot follow the question-conditioning prompt and regress. Multi-hop evidence fragments are simply ill-posed for pairwise judging. A small real top-k retrieval check over an embedded FEVER evidence corpus reproduces the same cost and over-flagging ladder on real retrieved pairs.

We package the findings as a five-row deployment ladder for vector-DB-backed RAG practitioners, with bootstrap confidence intervals throughout. Code and result artefacts are released.

REFERENCES

[1] Marc E. Canby, Xinchi Chen, Xing Niu, Jifan Chen, Bonan Min, Sergul Aydore, and Vittorio Castelli. 2025. Benchmarking Query-Conditioned Natural Language Inference. In *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics, Vienna, Austria, 14808–14835. https://doi.org/10.18653/v1/2025.findings-acl.765 Introduces query-conditioned NLI: the relation label is conditioned on a premise, hypothesis, AND a query specifying the aspect under comparison.

[2] Kaustubh Dhole, Kai Shu, and Eugene Agichtein. 2025. ConQRet: A New Benchmark for Fine-Grained Automatic Evaluation of Retrieval Augmented Computational Argumentation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, Albuquerque, New Mexico, 5687–5713. https://aclanthology.org/2025.naacl-long.293 Fine-grained retrieval evaluation for computational argumentation; stance-diversified retrieved evidence.

[3] Vignesh Gokul, Srikanth Tenneti, and Alwarappan Nakkiran. 2025. Contradiction Detection in RAG Systems: Evaluating LLMs as Context Validators for Improved Information Consistency. *arXiv preprint arXiv:2504.00180* (March 2025). https://arxiv.org/abs/2504.00180 Evaluates LLMs as context validators that scan retrieved context for contradictions, an LLM-as-judge baseline directly comparable to our llm_bare.

[4] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *ICML*. Temperature scaling, ECE.

[5] Mayank Ravishankara. 2025. FVA-RAG: Falsification-Verification Alignment for Mitigating Sycophantic Hallucinations. *arXiv preprint arXiv:2512.07015* (Dec. 2025). https://arxiv.org/abs/2512.07015

[6] Xinzhuo Sun, Jing Chen, Binghua Zhou, and Meng-Ju Kuo. 2024. ConRAG: Contradiction-Aware Retrieval-Augmented Generation under Multi-Source Conflicting Evidence. *Journal of Advanced Computing Systems* 4, 7 (2024). https://scipublication.com/index.php/JACS/article/view/257 Two-stage NLI-style passage tagging, consistent-evidence clustering, and answer generation with traceable citations.

[7] Chenyu Wang, Yueyuan Li, Yingmin Liu, and Yang Shu. 2026. ConflictRAG: Detecting and Resolving Knowledge Conflicts in Retrieval-Augmented Generation. *arXiv preprint arXiv:2605.17301* (2026). https://arxiv.org/abs/2605.17301 Two-stage conflict detection (lightweight embedding-based MLP classifier with selective LLM refinement) plus Entropy-TOPSIS source credibility and a Conflict-Aware RAG Score. Reports 90.8% detection accuracy, 88.7% conflict-detection F1, and 5.3–6.1% downstream correctness gains.

[8] Fei Wang, Xingchen Wan, Ruoxi Sun, Jiefeng Chen, and Sercan Ö. Arık. 2024. Astute RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts for Large Language Models. *arXiv preprint arXiv:2410.07176* (Oct. 2024). https://arxiv.org/abs/2410.07176

[9] Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. 2025. Retrieval-Augmented Generation with Conflicting Evidence. In *Conference on Language Modeling (COLM)*. Introduces the RAMDocs benchmark and the MADAM-RAG multi-agent debate; arXiv:2504.13079.

[10] Kevin Wu, Eric Wu, and James Zou. 2024. ClashEval: Quantifying the Tug-of-War between an LLM's Internal Prior and External Evidence. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. https://openreview.net/forum?id=WGoCZl2itU

[11] Linda Zeng, Rithwik Gupta, Divij Motwani, Yi Zhang, and Diji Yang. 2025. Worse than Zero-shot? A Fact-Checking Dataset for Evaluating the Robustness of RAG Against Misleading Retrievals. *arXiv preprint arXiv:2502.16101* (Feb. 2025). https://arxiv.org/abs/2502.16101